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Master in Management, Organisation and Business Economics

MASTER'S THESIS

INCENTIVES AND REGULATION OF ARTIFICIAL INTELLIGENCE SAFETY

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ABSTRACT

This thesis shows how an AI company can sell the same product in two versions when consumers do not share the same views on capability and safety. The model includes a standard and a premium version. The standard version is safer but has fewer capabilities, while the premium version offers more capabilities with a higher safety risk. Consumers decide which version to buy based on what matters more to them and on the prices set by the company.

The model differs from the usual product versioning approach because it does not consider quality in only one dimension. For an AI product, both capability and safety matter, and consumers may give them different importance. This makes it possible to study pricing and consumer choice while also accounting for AI safety.

One main result is that selling both versions can be more profitable than selling only one. The model also examines what happens when using the premium version can harm other people. If the company has to pay for part of this harm, this affects its prices and can make the premium version less attractive to consumers. However, increasing the firm’s liability does not necessarily produce the outcome that is best for society. Third-party harm is only one part of the problem. The firm’s monopoly position also affects consumer choices and the final allocation between the two versions.

The thesis ends with the case of xAI’s Grok, which is used to show how the trade-off between capability, safety and the firm’s incentives can also be seen in a real AI market.

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1. INTRODUCTION

In December 2024, xAI made Grok 2 available free of charge to X users, while paid subscribers continued to have higher usage limits and earlier access to new capabilities (xAI, 2024a). In July 2025, the company introduced Grok 4, along with the SuperGrok Heavy tier, which provided access to Grok 4 Heavy (xAI, 2025). As Grok became more widely available, concerns about some of its uses also increased. In January 2026, the California Attorney General opened an investigation into non-consensual sexually explicit material generated using Grok and later issued a cease-and-desist letter to xAI (California Department of Justice, 2026a, 2026b). This case provides a useful starting point for the question this thesis studies. AI firms can offer different versions of the same product, with different levels of access and restrictions, while some of the consequences of their use may fall on people who are not part of the transaction.

Artificial intelligence is increasingly offered as a commercial product rather than as a single technology with the same characteristics for every user. AI providers commonly offer several access tiers that differ in model capabilities, usage limits and available features. At the same time, AI systems also differ in the safeguards that restrict harmful or undesirable uses. These differences create a pricing problem that the standard idea of a single quality ladder does not capture well.

For a conventional digital product, a premium version is usually better along a dimension that all consumers value positively. AI products are more complicated. Consumers may all value both capabilities and safety, but not to the same degree. A casual user may place relatively more weight on safeguards and reliability, while a user who relies heavily on AI for demanding tasks may place greater weight on capabilities. As a result, two consumers can rank the same pair of AI products differently even when both value safety positively.

This thesis studies the incentives of a monopolist that sells two fixed versions of an AI system. The premium version has higher capabilities and lower safety, while the standard version has lower capabilities and higher safety. The firm takes these characteristics as given and chooses prices. Consumers differ in how much weight they place on capabilities relative to safety and select the version that gives them the highest utility.

The first question is a pricing question. Given the two products and heterogeneous consumer preferences, does the firm prefer to sell only one version or to offer both? The second question concerns social welfare. If access to the less safe premium version harms people outside the transaction, how does the firm's privately optimal allocation differ from the socially preferred one? The final question is whether liability can reduce this difference by making the firm bear part of the harm generated by premium users.

The main contribution of the thesis is to bring capability and safety into the standard versioning problem and then examine what happens when the choice of version also creates costs for third parties. The model builds on the usual self-selection mechanism, but allows products to differ in both capability and safety while consumers can still be placed along a single preference dimension. Safety directly affects utility and also affects usable capability, which determines how consumers sort between the two versions. Within this setting, the thesis derives analytical solutions for the single-version cases and for the interior two-version menu, and establishes the conditions under which offering both versions is more profitable. The analysis is then extended to third-party harm and liability. This makes it possible to compare the firm's preferred allocation with the social optimum and, importantly, to separate the externality's role from the monopoly distortion that exists even without third-party harm.

The model extends the economics of second-degree price discrimination, especially Belleflamme's (2005) versioning framework for information goods. It maintains the traditional self-selection mechanism but modifies consumer utility to differentiate between capabilities and safety. While safety influences utility directly, its marginal benefit decreases relative to the consumer type parameter. By defining usable capability, consumer sorting simplifies to a single dimension, despite the products still being characterized by two separate attributes.

The pricing analysis first considers the two single-version benchmarks. Under the maintained restrictions, utility from the standard version decreases with the consumer type, while utility from the premium version increases. Under the full-coverage condition, the standard version serves the entire market at its optimal benchmark price, whereas the premium version excludes consumers with low willingness to pay when the excluded-bottom condition holds. When both versions are offered and the interiority condition holds, the market is divided at a single cutoff. Consumers below the cutoff purchase the standard version and consumers above it purchase the premium version.

Under the conditions of the analytical comparison, the two-version menu generates weakly higher profit than the standard-only benchmark and strictly higher profit than the premium-only benchmark. Under the numerical example used in the thesis, the optimal cutoff is approximately $\theta^* = 0.735$, so about 26.5 percent of consumers purchase the premium version. These values are illustrative and are not estimates of actual behaviour on any AI platform.

The welfare analysis introduces expected harm generated by users of the premium version. Liability changes the firm's incentives because serving a premium user now carries an additional private cost. As the liability share increases, the firm raises the cutoff and shifts consumers toward the standard version. This reduces the size of the premium segment and lowers expected third-party harm.

Liability, however, does not generally make the firm's allocation equal to the social planner's allocation. Two separate reasons explain the difference. The first is the externality created by third-party harm. The second exists even when there is no external harm, because a monopolist maximizes profit rather than total surplus. In the baseline numerical example, the liability share required to reproduce the planner's allocation is $\lambda^* = 1.25$, which is outside the feasible range. The sensitivity analysis also shows that this result depends on the severity of harm. When harm is small, the monopoly distortion can dominate and additional liability can move the firm further from the social optimum. When harm is larger, stronger liability becomes more useful, but may still be insufficient.

The case of xAI's Grok is used to provide an empirical interpretation of the model. The case study does not attempt to estimate the model or establish a quantitative fit. Instead, it examines whether the mechanisms used in the analysis can be recognized in practice. Grok is useful for this purpose because xAI has combined free and paid access tiers with differences in capabilities and feature restrictions, while harmful uses of some Grok functions have also generated regulatory concern. Regulatory investigations following harmful deepfake generation illustrate why access decisions can affect people outside the consumer-firm relationship (California Department of Justice, 2026a, 2026b).

The remainder of the thesis is organized as follows. Chapter 2 reviews the literature on versioning, AI safety and liability. Chapter 3 presents the model, consumer preferences, product characteristics, the firm's objective and the timing of decisions. Chapter 4 solves the firm's pricing problem and compares the two-version menu with the single-version benchmarks. Chapter 5 introduces third-party harm, liability and social welfare, and studies the difference between the firm's choice and the planner's preferred allocation. Chapter 6 uses the case of Grok to interpret the main mechanisms in a real market setting. The final chapter concludes and discusses the limitations of the analysis and possible extensions.

2. LITERATURE REVIEW

Artificial intelligence raises economic questions that draw on several connected areas of research. AI systems are digital products, which makes the literature on versioning and second degree price discrimination a natural starting point. At the same time, safety creates problems that are not present in standard models of information goods. A less safe system can impose costs on people who are not involved in the transaction between the firm and the user. This connects the problem to the literature on externalities, product safety and liability. A further issue is whether capability and safety can be treated as separate product characteristics, which is important for the way the model in this thesis is constructed.
The purpose of this review is not to treat these literatures as separate topics. Rather, it identifies what each contributes to the question studied in the thesis. The versioning literature provides the basic mechanism through which heterogeneous consumers sort between products. The AI safety literature explains why firms' private incentives may not produce the socially preferred level of safety. The liability literature provides a way of studying how part of the external cost can be shifted back to the firm. Finally, recent developments in commercial AI motivate studying products that differ in both capability and safety.

2.1. VERSIONING AND SECOND-DEGREE PRICE DISCRIMINATION

The starting point is the literature on second-degree price discrimination. A monopolist may be unable to observe each consumer's willingness to pay directly, but it can still separate consumers by offering a menu of products. Consumers then reveal information about their preferences through the product they choose. This idea appears in Pigou's (1920) early analysis of price discrimination and was later developed formally in models of product quality, including Mussa and Rosen (1978).

Armstrong (2008) provides a broader treatment of price discrimination and its welfare effects. A central point is that greater ability to discriminate between consumers can increase the firm's profit, but the welfare effect is not necessarily positive or negative. It depends on factors such as market coverage, changes in output and the way surplus is distributed across consumers. This is relevant to the present thesis because offering an additional AI version can attract consumers who would otherwise not buy, but it can also change the allocation among consumers already in the market.

Versioning applies this logic to situations in which the firm offers different forms of the same underlying product. Shapiro and Varian (1998) discuss versioning in the context of information goods, where reproduction costs are low and product features can be used to separate consumers with different valuations. Jansen (2022) presents the same mechanism as a form of indirect price discrimination. Rather than observing consumer types directly, the firm offers different combinations of price and quality and lets consumers choose the option they prefer.

The closest theoretical foundation for the pricing part of this thesis is Belleflamme (2005). He studies a monopolist that can offer two versions of an information good to consumers who differ in their valuation of quality. Consumer utility is written as

$$U_i(\theta) = k + \theta s_i - p_i$$

Where k is a common valuation component, s_i represents product quality and θ captures heterogeneity in the value consumers place on quality. The firm cannot observe individual types and uses the price menu to induce self-selection.

Belleflamme shows that versioning creates two opposing effects. A lower-quality version can expand the market by attracting consumers who would not purchase the high-quality version, but it can also reduce profit through cannibalization when some consumers switch away from the high-quality product. Whether versioning is profitable depends on the balance between these effects.

This mechanism provides the main economic benchmark for Chapter 4 of this thesis. The present model also compares the profit from a two-version menu with the profits from offering each version separately. The difference is that the two AI products are characterized by both capability and safety, even though consumer sorting remains one-dimensional. In the current model, the premium version has greater capability but lower safety.

This creates a different source of consumer heterogeneity from the usual interpretation of product quality. A consumer who places greater weight on capability may prefer the premium version despite its lower safety, while another consumer may prefer the safer standard version. The sorting mechanism remains similar to that used in traditional versioning models, but the interpretation of product characteristics differs.

Deneckere and McAfee (1996) provide a related argument in their analysis of damaged goods. They show that a firm may benefit from offering a deliberately degraded product when doing so helps separate consumers with different valuations. Their mechanism is not the same as the one used in this thesis, since the characteristics of the two AI products here are taken as given rather than chosen by the firm. Still, their analysis is useful for understanding why differences between versions can have economic value even when one version is inferior along an important product characteristic.

The versioning literature therefore provides the main tools for analysing pricing and consumer sorting. Its limitation for the question studied here is that product differences normally affect the firm and its customers. AI safety introduces another group whose welfare can also be affected, namely third parties who do not participate in the transaction.

2.2. AI SAFETY AND COMPETITIVE INCENTIVES

A separate literature studies the incentives firms face when making decisions about AI safety. The main concern in this literature is that the private benefits from developing or deploying a system may not reflect all of the social risks created by that system.

Choi, Jeon and Menicucci (2026) study competition in the timing of AI deployment when deployment involves safety risk. They identify two possible distortions. A first-mover advantage can encourage firms to deploy too early, creating a race to the bottom in safety. At the same time, firms may delay deployment because they can learn from competitors' experimentation. Their analysis shows that the relationship between competition and AI safety is not straightforward and depends on the incentives created by the market environment.

Bueno de Mesquita, Dziuda and Polborn (2026) consider a related problem in the context of an artificial general intelligence race. Firms allocate resources between speed and safety. Resources devoted to speed improve the probability of reaching AGI first, while resources devoted to safety reduce the risk of a catastrophic outcome. Their analysis shows that market structure can affect how firms divide resources between these two objectives and, as a result, can affect the level of risk.

These papers approach AI safety from a different direction than the present thesis. They focus on competition between developers, resource allocation or the timing of deployment. The model developed here does not consider firms competing to develop AGI, nor does it analyze the timing of deployment. Instead, it assumes that a monopolist already has access to two AI products with given characteristics and asks how those products are priced.

The connection between the two approaches is incentives. The AI safety literature shows no general reason to expect private decisions about safety to coincide with socially preferred outcomes. The present thesis applies this idea to a pricing and versioning problem. Once one of the available products can generate harm outside the transaction, the firm's decision about how consumers are allocated between versions also affects social welfare.

2.3. PRODUCT SAFETY AND LIABILITY

The possibility of harm to third parties connects the model to the economic literature on product safety and liability. The basic problem is that a producer may not bear the full cost of the harm its product creates. When this happens, private incentives can differ from social incentives.

Shavell (1984) studies the use of liability and safety regulation as instruments for controlling accident risk. Liability changes incentives by making the producer bear some or all of the losses associated with harmful outcomes. Safety regulation can instead impose requirements directly. The relative usefulness of these instruments depends on the environment and on the information available to firms, consumers and regulators.

More recent work applies similar questions directly to artificial intelligence. Chen and Hua (2026) study product safety when firms choose the autonomy of AI-powered products and make research and development decisions. Their analysis shows that liability can improve safety incentives by making firms internalize more of the expected harm their products cause. At the same time, liability does not necessarily reproduce the fully efficient outcome in every setting.

The mechanism in this thesis is different. The firm does not choose AI autonomy or research effort. It takes the characteristics of the two versions as given and chooses prices. Liability affects the firm's incentives because users of the premium version generate expected third-party harm. When the firm is responsible for a share of that harm, serving consumers through the premium version becomes more costly.

This creates an important distinction between reducing an externality and achieving the social optimum. Liability can address the externality created by third-party harm, but the firm remains a monopolist and continues to maximize profit rather than total welfare. This suggests that reducing the externality and reproducing the planner's preferred allocation need not be the same objective.

This distinction motivates the analysis in Chapter 5. The question is not only whether liability reduces harmful use, but also how it changes the allocation between the two versions and whether the resulting allocation moves closer to the social optimum. The analysis also considers whether this relationship changes when the externality is relatively small.

2.4. CAPABILITIES, SAFETY AND EMPIRICAL MOTIVATION

The model requires a product configuration in which the premium version has greater capability but lower safety than the standard version. This assumption should not be interpreted as a general technological relationship between capability and safety. A more capable AI system does not necessarily have to be less safe.

Rather, the model considers a situation in which the available premium version has higher capability and lower safety than the standard version. The model does not explain how this product configuration is created. It takes the characteristics of the two versions as given and studies the firm's pricing decision once these products are available.

This distinction is important because capability and safety are conceptually different characteristics. Capability refers to what the system is able to do, while safety refers to the restrictions and safeguards that affect how those capabilities can be used. In practice, AI providers can change access conditions, content restrictions and safeguards without necessarily changing the underlying technical capability of the model.

The Grok case serves as a helpful example. xAI has built Grok across multiple model and subscription tiers, offering free access as well as paid plans that provide extra features. Later, the company launched more sophisticated models, such as Grok 4 and Grok 4 Heavy, with access levels varying by subscription (xAI, 2024b; xAI, 2025).

The Grok case is also relevant because the product has been associated with a more permissive approach to some types of content than competing AI systems (xAI, 2023). This does not establish a precise measure of safety and it does not imply that every paid Grok product is less safe than every free product. It does, however, provide a useful setting in which capability, access and restrictions are not necessarily moving together.

The issue of third-party harm highlights another reason to examine Grok. In January 2026, the California Attorney General launched an investigation into sexually explicit non-consensual content created with Grok and sent a cease-and-desist order to xAI (California Department of Justice, 2026a, 2026b). This situation demonstrates how employing an AI tool can impose costs on people outside the direct provider-user interaction.

The empirical case is used only as an illustration. The Grok data are not used to estimate the parameters of the model, and the numerical values used later in the thesis should not be interpreted as estimates of xAI's actual consumers, safety levels or liability exposure. The purpose of the case study is instead to examine whether the economic mechanisms described by the model have a recognizable counterpart in a real AI market.

2.5. LITERATURE GAP AND CONTRIBUTION

The literature discussed above provides different parts of the framework needed for the present analysis. The versioning literature explains how a monopolist can use prices and product differences to induce self-selection among heterogeneous consumers. The AI safety literature shows how private incentives can differ from socially preferred safety outcomes. The liability literature provides a way to study what happens when part of the cost of harm is shifted back to the firm.

These questions are usually studied separately. Standard versioning models do not normally consider a product characteristic that can generate harm for third parties. AI safety models, on the other hand, generally focus on development races, deployment decisions, research effort or the level of safety chosen by firms rather than on the pricing of different AI versions.

The model in this thesis brings these elements together in a simple setting. It considers a monopolist offering two AI products. The premium product has higher capability but lower safety, while the standard product has lower capability but higher safety. Consumers differ in how strongly they value capability relative to safety. Higher values of the type parameter increase the weight placed on capability and reduce, but do not eliminate, the weight placed on safety. A consumer who places relatively more weight on safety can therefore prefer the standard product even when the premium product has greater technical capability.

I do not attempt to solve a fully multidimensional screening problem. Consumer heterogeneity remains one-dimensional, and the transformation to usable capability preserves the single crossing structure used in standard versioning models. The products themselves, however, differ in two characteristics that have different interpretations in consumer utility. Safety also enters utility directly, rather than appearing only as part of the quality index.

The introduction of third-party harm adds a second element to the analysis. Pricing determines how consumers are divided between the two versions, and this allocation determines the amount of expected harm generated by premium users. Liability can therefore affect the allocation indirectly by changing the firm's private cost of serving the premium segment.

This raises a further question about the relationship between liability and social welfare. Reducing third-party harm does not necessarily imply that the firm's allocation will coincide with the allocation preferred by a social planner. The firm continues to maximise profit, while the planner also takes consumer surplus into account. The analysis in Chapter 5 examines these two sources of divergence separately and asks how far liability can move the firm's choice toward the social optimum.

The thesis's contribution is therefore relatively focused. It does not attempt to provide a general model of AI safety or to explain how firms choose the technology behind an AI system. Instead, it takes product characteristics as given and studies what happens when a monopolist prices two versions that differ in both capability and safety, and when the less safe version can harm third parties. This leads to the main research question of the thesis:

How does a monopolist's pricing influence the distribution between a safer standard version and a more capable, yet less safe premium version when consumers vary in

their valuation of AI capability and safety? Additionally, to what degree can liability mitigate the welfare distortion caused by this?

3. THE MODEL

This chapter develops the thesis's theoretical framework. We model a monopolist that sells an AI system in two versions to a heterogeneous population of consumers. The firm chooses prices for both versions; consumers choose which version to buy, or whether to buy at all.

The key feature that distinguishes this model from standard versioning frameworks is that product quality has two dimensions - capabilities and safety that consumers may value differently. A power user who relies on AI for complex tasks may place high value on capabilities and relatively little on safety restrictions. A casual user who uses AI occasionally may value safety more. This heterogeneity allows consumers to value the same product characteristics differently and creates the basis for self-selection between versions.

3.1. CONSUMERS

Not all users of an AI system want the same thing, and this heterogeneity is the model's starting point. We consider a unit mass of consumers indexed by a preference parameter θ, uniformly distributed on [0, 1]. Low values of θ correspond to casual users, while high values correspond to power users. Casual users use AI mainly for simpler tasks and attach greater value to safety, while power users rely more heavily on capabilities (c_i) and place relatively less weight on safety (s_i). Consumer utility from purchasing version i at price p_i is

$$U_i(\theta) = \theta c_i + (1 - \alpha\theta)si - p_i,$$

$$\alpha \epsilon (0,1) \text{ (1)}$$

The outside option is normalized to zero.

The marginal value of capabilities is

$$\frac{\partial U_i}{\partial c_i} = \theta \geq 0.$$

All consumers therefore value higher capabilities, but the value of an increase in capabilities is larger for consumers with higher θ.

The marginal value of safety is

$$\frac{\partial U_i}{\partial s_i} = 1 - \alpha\theta$$

Because $\alpha \in (0,1)$ and $\theta \in [0,1]$ this expression is strictly positive for every consumer type. Safety therefore increases utility for all consumers. However, its marginal value decreases with θ. A consumer with $\theta = 0$ places weight 1 on safety, while a consumer with $\theta = 1$ places weight $1 - \alpha$ on it. The parameter α captures how strongly consumers who value capabilities place less weight on safety.

We define usable capability as

$$q_i \equiv c_i - \alpha si$$

Utility can then be written as

$$U_i(\theta) = s_i + \theta q_i - p_i$$

(2)

This form makes the self-selection mechanism easier to see. The cross-partial derivative is

$$\frac{\partial^2 U_i}{\partial\theta\partial q_i} = 1 > 0$$

This gives the model the single-crossing property: consumers with higher values of θ value increases in usable capability more strongly. This allows the standard self-selection logic from the versioning literature, including Belleflamme (2005), to be applied without the firm observing consumer types directly. At the same time, consumer sorting depends on the single index q_i, while safety continues to affect utility directly through the intercept s_i. The product space is therefore two-dimensional, even though consumers can be ordered along the single preference dimension θ.

The intercept s_i plays an important role in the model. In equation (2), safety affects utility both directly through s_i and indirectly through usable capability $q_i = c_i - \alpha si$. The direct term s_i enters utility independently of θ and varies across product versions. In Belleflamme's standard quality-screening framework, the corresponding intercept is common across versions and therefore does not affect consumers' choice between them. Here, by contrast, the intercept is version-specific because safety differs across products.

This difference matters particularly for the standard version. If safety entered utility only through q_i, a consumer with $\theta = 0$ would obtain

$$U_i(0) = -p_i.$$

At any positive price, this consumer would therefore choose the outside option. In the present specification, the value obtained by a consumer with $\theta = 0$ therefore comes entirely from safety. This can be interpreted as the value of guardrails in providing trust and protection against unwanted outputs. The version-specific safety term s_i gives the standard version a direct source of utility for low-θ consumers, independently of the value they obtain from usable capability. This feature distinguishes the present model from the standard one-dimensional quality-screening framework.

3.2. PRODUCTS

The previous section explained how consumers differ in their preferences. The next step is to define the products they choose between.

The firm offers two versions, denoted by H and L. Each version is described by two characteristics: capabilities $c_i \in [0,1]$ and safety $s_i \in [0,1]$. Each version is therefore a fixed pair

$$(c_i, s_i) \in [0,1]^2$$

The firm takes these product characteristics as given.

Capabilities measure the model's ability to perform complex tasks, such as advanced reasoning, code generation, or specialised domain knowledge. Higher values of c_i represent better performance. Safety measures how well the model avoids harmful outputs and refuses unsafe requests. Higher values of s_i represent stronger safety measures, while lower values indicate fewer guardrails.

As defined in Section 3.1, usable capability is

$$q_i = c_i - \alpha s_i$$

We also define the difference in usable capability between the two versions as

$$\Delta q \equiv q_H - q_L = \Delta c + \alpha \Delta s$$

The maintained restrictions on the product characteristics and consumer preferences are

$$\Delta c \equiv c_H - c_L > 0, \Delta s \equiv s_L - s_H > 0, \qquad \frac{c_L}{s_L} < \alpha < \frac{c_H}{s_H} \quad (3)$$

The first two restrictions describe the product configuration considered in the analysis: version H has higher capabilities and lower safety than version L. The third restriction requires α to lie strictly between the capability-to-safety ratios of the two versions. It is equivalent to

$$q_L < 0 < q_H$$

This makes clear that the sign of usable capability is a parameter restriction rather than a result of the model. It also requires version H to have a higher capability-to-safety ratio than version L.

Under the example used later in the numerical analysis, the restriction in (3) becomes

$$0.625 < 0.8 < 5$$

and is therefore satisfied.[1] The feasibility of this configuration is taken as given; the empirical evidence discussed in Section 2.4 is what makes it plausible.

3.3. THE FIRM

The firm is a monopolist. It takes the product characteristics (c_L, s_L, c_H, s_H), as given and chooses only the prices (p_L, p_H).

Marginal cost is normalized to zero.

Given prices, each consumer chooses the version that gives the highest utility, or takes the outside option if neither version provides positive utility. Let n_L and n_H denote the mass of consumers who purchase versions L and H, respectively. The firm's profit is therefore

$\pi \ = \ p_L n_L + p_H n_H$. (4)

[1]Chapter 4 will additionally require $s_L + 2q_L \geq 0$. for the full-coverage standard-version benchmark. This condition ensures that serving the entire market is profit-maximizing. Since $q_L < 0$, it also implies $s_L + q_L > 0$, so the resulting full-coverage price is positive. Under the numerical example, $s_L + 2q_L = 0.52 \geq 0$.

The firm's objective is to choose prices to maximize profit, taking consumer self-selection into account. Chapter 4 analyzes the resulting pricing problem.

3.4. TIMING

The model has a simple sequential structure**.** The product characteristics (c_L, s_L, c_H, s_H) are fixed and known to both the firm and consumers. Consumers make their choices after observing the prices. The timing is as follows.

Stage 0. The firm chooses which versions to offer. It may offer only version L, only version H, or both versions.

Stage 1. The firm observes the product characteristics and sets the price or prices for the version or versions offered in Stage 0.

Stage 2. Consumers observe the available versions and their prices. Each consumer chooses version L, version H, or the outside option, depending on which gives the highest utility.

The model is solved by backward induction. Consumer choices at Stage 2 determine demand for each version. Anticipating these choices, the firm chooses prices at Stage 1 to maximize profit. At Stage 0, the firm chooses the offering that generates the highest profit. Chapter 4 derives the optimal pricing decisions and compares the two-version menu with the single-version benchmarks.

3.5. RELATION TO STANDARD VERSIONING MODELS

The model retains the main self-selection mechanism of standard versioning models such as Belleflamme (2005). Since

$$\frac{\partial^2 U_i}{\partial \theta \partial q_i} = 1 > 0,$$

consumers can be ordered by a single preference parameter θ, with higher-θ consumers placing greater value on usable capability q_i. This single-crossing property allows the standard sorting logic to carry over to the present setting. The transformation

$$q_i = c_i - \alpha s_i$$

therefore reduces consumer sorting to a one-dimensional problem.

However, the transformation does not eliminate the separate role of safety. In the reduced utility function

$$U_i(\theta) = s_i + \theta q_i - p_i,$$

safety also enters directly through the version-specific intercept s_i. In Belleflamme's framework, the corresponding intercept is common across versions, whereas here it varies with the safety level of each product. The model is therefore not fully two-dimensional in the screening sense: consumers can still be ordered along a single dimension. However, the products themselves remain two-dimensional because capabilities and safety enter utility in different ways. This preserves the standard self-selection structure while allowing safety to play a separate role in consumers' valuation of each version.

4. THE FIRM'S PROBLEM

The firm chooses prices taking the product characteristics and consumer preferences as given. Its pricing decision determines which consumers purchase each version and whether some consumers choose the outside option.

We solve the problem in two steps. We first consider the two single-version benchmarks, where only L or only H is available. We then solve the two-version menu and compare its profitability with the benchmark outcomes. The numerical example at the end of the chapter illustrates the solution under the parameter values used throughout the thesis.

4.1. SINGLE-VERSION BENCHMARKS

We first consider the firm's pricing problem when only one version is available. We analyze version L first and then version H. These outcomes provide the benchmarks for the two-version menu considered in Section 4.2.

Only the standard version (L)

We consider a firm that offers only the standard version, L. The firm's objective is to choose the price p_L that maximizes profit while ensuring that consumers are willing to purchase the product. Under the maintained restriction $q_L < 0$, utility from version L is decreasing in θ, the utility from version L is

$$U_L(\theta) = s_L + \theta q_L - p_L.$$

Since $q_L < 0$, utility is decreasing in θ:

$$\frac{\partial U_L}{\partial \theta} = q_L < 0.$$

Since $q_L < 0$, consumers with higher values of θ obtain lower utility from version L. The participation constraint is therefore determined by the highest consumer type, $\theta = 1$, who values version L the least.

Let $\hat{\theta}_L$ denote the marginal consumer who is indifferent between purchasing version L and choosing the outside option. Since utility from L is decreasing in θ, consumers in the interval $[0, \hat{\theta}_L]$ purchase the product. The participation constraint of the marginal consumer is

$$U_L(\hat{\theta}_L) = s_L + \hat{\theta}_L\, q_L - p_L = 0,$$

which gives $p_L(\hat{\theta}_L) = s_L + q_L\hat{\theta}_L$.

Because q_L<0, the price is decreasing in $\hat{\theta}_L$. Keeping consumers with higher values of θ in the market therefore requires the firm to charge a lower price. Profit as a function of the marginal consumer is $\pi_L\ (\hat{\theta}_L) = (s_L + q_L\hat{\theta}_L)\hat{\theta}_L$.

Differentiating with respect to $\hat{\theta}_L$ gives

$$\frac{d\pi_L}{d\hat{\theta}_L} = s_L + 2q_L\hat{\theta}_L, \text{ and}$$

$$\frac{d^2\pi_L}{d^2\hat{\theta}_L} = 2q_L < 0.$$

Profit is therefore concave in $\hat{\theta}_L$. Full market coverage corresponds to the upper corner $\hat{\theta}_L = 1$. This corner is optimal if and only if

$$\left.\frac{d\pi_L}{d\hat{\theta}_L}\right|_{\hat{\theta}_L = 1} = s_L + 2q_L \geq 0.$$

The economics of this condition can be seen by considering a small reduction in coverage from $\hat{\theta}_L$=1. Excluding a marginal group of the highest consumer types allows the firm to increase the price charged to all remaining consumers, since $q_L < 0$. At the same time, the firm loses the revenue generated by the consumers who leave the market. At full coverage, the revenue lost from excluding a marginal consumer is $s_L + q_L$, while the marginal price increase on the consumers retained is $-q_L$. Full coverage is therefore optimal when

$$s_L + q_L \geq -q_L, \text{ or equivalently,}$$

$$s_L + 2q_L \geq 0.$$

This condition is stronger than $s_L + q_L > 0$, which only ensures that the full-coverage price is positive. The condition $s_L + 2q_L \geq 0$ additionally ensures that full coverage is profit-maximizing. When $s_L + 2q_L \geq 0$, the optimal marginal consumer is $\hat{\theta}_L{}^* = 1$. The optimal price is therefore

$$p_L{}^* = s_L + q_L, \text{ the market is fully covered, and profit is}$$

$$\pi_L{}^* = s_L + q_L.$$

If instead $s_L + 2q_L < 0$, the full-coverage corner is no longer optimal. The first-order condition gives an interior marginal consumer $\hat{\theta}_L{}^* = -\frac{s_L}{2q_L}$, so the firm serves only consumers in $[0, \hat{\theta}_L{}^*]$. The corresponding optimal price is

$$p_L{}^* = \frac{s_L}{2}, \text{ and profit is}$$

$$\pi_L{}^* = \frac{s_L{}^2}{-4q_L}.$$

Only the premium version (H)

Consider now a firm that offers only the premium version, H. Under the maintained restriction $q_H > 0$, utility from version H is increasing in θ:

$$U_H(\theta) = s_H + \theta q_H - p_H.$$

Since $q_H > 0$, utility is increasing in θ:

$$\frac{dU_H}{d\theta} = q_H > 0.$$

Consumers with higher values of θ value version H the most, while consumers with lower values of θ value it the least. The consumer most likely to leave the market is therefore the lowest consumer type, $\theta = 0$, whose utility is

$$U_H(0) = s_H - p_H,$$

which becomes negative whenever $p_H > s_H$.

Let $\hat{\theta}_H$ denote the marginal consumer, who is indifferent between purchasing version H and choosing the outside option. Since utility from H is increasing in θ, consumers in the interval $[\hat{\theta}_H, 1]$ purchase the product. The participation constraint of the marginal consumer is

$$U_H(\hat{\theta}_H) = s_H + \hat{\theta}_H q_H - p_H = 0,$$

which gives

$$p_H(\hat{\theta}_H) = s_H + q_H\hat{\theta}_H.$$

Because $q_H > 0$, the price is increasing in $\hat{\theta}_H$. Excluding more consumers with low values of θ allows the firm to charge a higher price to the consumers who remain. Profit is therefore

$$\pi_H\left(\hat{\theta}_H\right) = (s_H + q_H\hat{\theta}_H)(1 - \hat{\theta}_H)$$

Differentiating gives

$$\frac{d\pi_H}{d\hat{\theta}_H} = q_H - s_H - 2q_H\hat{\theta}_H, \text{ and}$$

$$\frac{d^2\pi_H}{d^2\hat{\theta}_H} = -2q_H < 0.$$

Profit is again concave. Full market coverage corresponds to the lower corner $\hat{\theta}_H = 0$. Excluding consumers from the bottom of the market is profitable only if profit increases when the firm moves away from this corner:

$$\left.\frac{d\pi_H}{d\hat{\theta}_H}\right|_{\hat{\theta}_H = 0} = q_H - s_H > 0.$$

The excluded-bottom solution therefore requires $q_H > s_H$. Using $q_H \equiv c_H - \alpha s_H$, this condition is equivalent to $c_H > (1 + \alpha)s_H$.

Economically, excluding a marginal group of low consumer types allows the firm to increase the price charged to all remaining consumers by q_H , while it loses the revenue s_H generated by the marginal consumer at full coverage. Exclusion is therefore profitable precisely when

$q_H > s_H$. When this condition holds, the optimal marginal consumer is interior. To keep the notation used below, let $\theta_0 \equiv \hat{\theta}_H$.

The first-order condition is

$q_H - s_H - 2q_H\theta_0 = 0$, which gives:

$$\theta_0^* = \frac{q_H - s_H}{2q_H},$$

$$\pi_H^* = \frac{(q_H + s_H)^2}{4q_H}.$$

$$p_H^* = \frac{q_H + s_H}{2}$$

If instead $q_H \leq s_H$, the lower corner $\hat{\theta}_H = 0$ is optimal. The firm then covers the entire market with version H, charges $p_H^* = s_H$, and earns $\pi_H^* = s_H$.

Proposition 1 (Single-Version Benchmarks):

Suppose $\mathrm{q_L} < 0 < \mathrm{q_H}$.

1. **Standard version:**

If $\mathrm{s_L} + 2\mathrm{q_L} \geq 0$, the optimal standard-version benchmark has full market coverage:

$\hat{\boldsymbol{\theta}}_{\boldsymbol{L}}^* = \mathbf{1}$, with optimal price

$\mathbf{p_L^*} = \mathbf{s_L} + \mathbf{q_L}$, and profit

$\boldsymbol{\pi}_{\mathbf{L}}^* = \mathbf{s_L} + \mathbf{q_L}$.

If instead $\mathrm{s_L} + 2\mathrm{q_L} < 0$, the firm excludes consumers with the highest values of θ. The marginal consumer is:

$\hat{\boldsymbol{\theta}}_{\boldsymbol{L}}^* = \frac{-\boldsymbol{s_L}}{\mathbf{2}\boldsymbol{q_L}}$, the optimal price is

$\mathbf{p_L^*} = \frac{\mathbf{s_L}}{\mathbf{2}}$, and profit is

$\boldsymbol{\pi}_{\mathbf{L}}^* = -\frac{\mathbf{s^2}_{\mathbf{L}}}{\mathbf{4}\boldsymbol{q_L}}$.

Consumers with $\theta > \hat{\theta}_L^*$ choose the outside option.

2. **Premium version:**

If $q_H > s_H$, equivalently $c_H > (1+\alpha)s_H$, the firm excludes consumers with the lowest values of θ. The marginal consumer is

$$\boldsymbol{\hat{\theta}_H^* = \theta_0^* = \frac{q_H - s_H}{2q_H}},$$ the optimal price is

$$\mathbf{p_H^*} = \frac{\mathbf{q_H + s_H}}{\mathbf{2}},$$ and profit is

$$\boldsymbol{\pi_H^*} = \frac{\mathbf{(q_H + s_H)^2}}{\mathbf{4q_H}}.$$

Consumers with $\theta < \theta_0^*$ choose the outside option.

If instead $q_H \le s_H$, the premium-version benchmark has full market coverage:

$$\boldsymbol{\hat{\theta}_H^* = 0},$$ with

$$\mathbf{p_H^*} = \boldsymbol{s_H}$$

$$\boldsymbol{\pi_H^*} = \boldsymbol{s_H}.$$

The two benchmark outcomes depend on the conditions stated in Proposition 1. For version L, full coverage is optimal when $s_L + 2q_L \ge 0$. In this case, the revenue lost by excluding consumers with high values of θ is greater than the gain from raising the price charged to the consumers who remain. For version H, the firm excludes consumers with low values of θ when $q_H > s_H$, because the higher price obtained from the remaining consumers more than compensates for the lost demand.

The numerical example used in this thesis satisfies both of these conditions. The relevant benchmarks for the analysis are therefore full market coverage for version L and an excluded-bottom solution for version H. These outcomes provide the benchmarks for the two-version menu considered in Section 4.2.

4.2. THE TWO-VERSION MENU

We restrict attention to two-version menus that cover the market, so every consumer purchases either version L or version H. Under the benchmark conditions $s_L + 2q_L \ge 0$ and

$q_H > s_H$, this restriction can also be justified from the more general problem in which an interval of consumers may choose the outside option.[2]

Since $q_H - q_L = \Delta q > 0$, the utility difference $U_H(\theta) - U_L(\theta)$ is strictly increasing in θ. Consumer preferences therefore satisfy single crossing, so there can be at most one consumer who is indifferent between the two versions. For an interior menu, let this consumer be denoted by θ_{HL}. Consumers below the cutoff choose version L, while consumers above it choose version H.

The consumer at θ_{HL} is indifferent between the two versions, so

$$U_L\ (\theta_{HL})\ = U_H\ (\theta_{HL}).$$

Because utility from L decreases with θ and utility from H increases with θ, the cutoff consumer receives the lowest utility among consumers who purchase. Under full market coverage, the participation constraint must therefore bind at this type. To see why, suppose instead that

$$U_L\ (\theta_{HL}) = U_H\ (\theta_{HL}) > 0.$$

The firm could then increase both $p_L$and p_H by the same small amount $\varepsilon > 0$. Utility from both versions would fall by the same amount, so the difference $U_H\ (\theta) - U_L\ (\theta)$ would remain unchanged. Consumer sorting and the cutoff would therefore be unaffected, while profit would increase. The firm continues to raise both prices until

$$U_L\ (\theta_{HL}) = U_H\ (\theta_{HL}) = 0$$

The equality between U_L and U_H is the indifference condition that defines the cutoff, while the zero-utility condition is the single binding participation constraint.

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[2] To see this, allow the highest type purchasing L and the lowest type purchasing H to differ, and denote them by $\hat{\theta}_L$ and $\hat{\theta}_H$, with $\hat{\theta}_L \leq \hat{\theta}_H$. If the inequality is strict, consumers in $(\hat{\theta}_L, \hat{\theta}_H)$ choose the outside option. Prices are then $p_L = s_L + q_L\hat{\theta}_L$ and $p_H = s_H + q_H\hat{\theta}_H$ and profit is $\boldsymbol{\Pi}(\hat{\boldsymbol{\theta}}_L, \hat{\boldsymbol{\theta}}_H) = (\boldsymbol{s_L} + \boldsymbol{q_L}\hat{\boldsymbol{\theta}}_L\ )\hat{\boldsymbol{\theta}}_L + (\boldsymbol{s_H} + \boldsymbol{q_H}\hat{\boldsymbol{\theta}}_H\ )(\mathbf{1} - \hat{\boldsymbol{\theta}}_H)$. Ignoring the constraint $\hat{\theta}_L \leq \hat{\theta}_H$, the two terms correspond to the single-version problems derived in Section 4.1. Under $s_L + 2q_L \geq 0$, the standard-version problem has $\hat{\boldsymbol{\theta}}_L^{\ *} = 1$. Under $q_H > s_H$, the premium-version problem has $\hat{\boldsymbol{\theta}}_H^{\ *} = \frac{q_H - s_H}{2q_H} < 1/2$. The unconstrained optima therefore violate $\hat{\theta}_L \leq \hat{\theta}_H$. The constraint must bind, so $\hat{\theta}_L = \hat{\theta}_H$, implying that no interval of consumers is excluded.

Reducing the optimization problem

Once the cutoff consumer is determined, the indifference condition and the binding participation constraint determine both prices. Solving the two equations above gives

$$p_L(\theta_{HL}) = s_L + \theta_{HL} q_L,$$
$$p_H(\theta_{HL}) = s_H + \theta_{HL} q_H.$$

The firm's optimization problem therefore depends on a single decision variable, the location of the cutoff consumer θ_{HL}, rather than on two independent prices.

The relationship between the cutoff and prices has a straightforward economic interpretation. A higher cutoff means that more consumers purchase version L, including consumers who value it less. The firm must therefore lower the price of the standard version to keep these consumers in the market. At the same time, the remaining buyers of version H have stronger preferences for capabilities, allowing the firm to charge a higher price for the premium version.

Profit can now be written as

$$\pi(\theta_{HL}) = p_L(\theta_{HL})\theta_{HL} + p_H(\theta_{HL})(1 - \theta_{HL}).$$

Substituting the pricing rules yields

$$\pi = s_H + (s_L - s_H + q_H)\theta_{HL} + (q_L - q_H)\theta_{HL}^2.$$

The optimal solution

Profit is a concave quadratic function of θ_{HL}because $q_L - q_H < 0$. The first-order condition therefore yields a unique optimum:

$$\theta_{HL}^* = \frac{s_L - s_H + q_H}{2\Delta q} = \frac{\Delta s + q_H}{2\Delta q}.$$

For the solution to be interior, the optimal cutoff must satisfy $0 < \theta^*_{HL} < 1$. The lower bound follows from $\Delta s > 0,\ q_H > 0, and\ \Delta q > 0$. The upper bound requires

$$\Delta s + q_H < 2\Delta q.$$

We maintain this additional condition for the interior two-version solution considered below.

Substituting this cutoff back into the pricing equations gives the complete solution:

$$\begin{aligned} p^*_L &= s_L + \theta^*_{HL} q_L, \\ p^*_H &= s_H + \theta^*_{HL} q_H, \\ n^*_L &= \theta^*_{HL},\ n^*_H = 1 - \theta^*_{HL}, \\ \pi^* &= p^*_L n^*_L + p^*_H n^*_H. \end{aligned}$$

Proposition 2 (Optimal Two-Version Menu):

Suppose the maintained restrictions of Section 3.2 hold, $\boldsymbol{s_L + q_L > 0}$, and $\Delta s + q_H < 2\Delta q$. The firm's optimal versioning menu is then characterised by:

1. the interior cutoff $\boldsymbol{\theta^*_{HL} = \frac{\Delta s + q_H}{2\Delta q}}$,
2. prices $\boldsymbol{p^*_L = s_L + \theta^*_{HL} \cdot q_L}$ and $\boldsymbol{p^*_H = s_H + \theta^*_{HL} \cdot q_H}$,
3. market shares $\boldsymbol{n^*_L = \theta^*_{HL}}$ and $\boldsymbol{n^*_H = 1 - \theta^*_{HL}}$
4. profit $\boldsymbol{\pi^* = p^*_L \cdot n^*_L + p^*_H \cdot n^*_H}$.

At the optimal cutoff, the consumer is indifferent between versions L and H, and the participation constraint binds at that same type.

4.3. THE PROFITABILITY OF VERSIONING

Comparing the menu with the standard version (L)

The two-version problem always generates weakly higher profit than the standard-version benchmark because the single-version L outcome can be replicated as a boundary case. When the optimal cutoff is interior, the firm additionally serves higher-θ consumers through version H and earns strictly more than under the L-only benchmark.

Economically, this result reflects the role of versioning in market segmentation. Consumers with lower values of θ continue to purchase the standard version, while consumers with higher values of θ voluntarily switch to the premium version. The firm is therefore able to earn additional profit from consumers with a higher willingness to pay for capabilities while preserving demand for the standard version.

Proposition 3 (Profitability of Versioning). Under the full-coverage condition $\boldsymbol{s_L + 2q_L \geq 0}$, the excluded bottom condition $q_H > s_H$, and the conditions of Propositions 1 and 2, the optimal two-version menu generates weakly higher profit than the L-only benchmark and strictly higher profit than the H-only benchmark. The L-only outcome can be replicated by setting $\boldsymbol{\theta_{HL} = 1}$.

Comparing the menu with the premium version (H)

The same logic applies when comparing the menu with the H-only benchmark. Under the condition $q_H > s_H$, let

$$\boldsymbol{\theta_0^* = \frac{q_H - s_H}{2q_H}}$$

denote the optimal cutoff when only version H is offered. Consider a two-version menu with $\boldsymbol{\theta_{HL} = \theta_0^*}$ **.** The premium price is then $\boldsymbol{p_H = s_H + q_H{}^*\theta_0^* = \frac{q_H + s_H}{2}}$, which is exactly the optimal H-only price. Consumers in $[\boldsymbol{\theta_0^*, 1}]$ therefore purchase version H on exactly the same terms as in the H-only benchmark. The menu additionally sells version L to consumers in $[\boldsymbol{0, \theta_0^*}]$**.** Its profit at this cutoff is

$$\boldsymbol{\pi(\theta_0^*) = \pi_H^* + (s_L + q_L\theta_0^*)\theta_0^*.}$$

Since $\boldsymbol{q_L < 0, \theta_0^* < 1}$ and $\boldsymbol{s_L + q_L > 0,}$ we have

$$\boldsymbol{s_L + q_L\theta_0^* > s_L + q_L > 0.}$$

The additional revenue from the standard segment is therefore strictly positive. Hence

$$\boldsymbol{\pi(\theta_0^*) > \pi_H^*.}$$

Since the optimal two-version menu earns at least as much as this feasible menu, the optimal menu strictly dominates the H-only benchmark.

Economic interpretation

The benchmark comparisons highlight the economic role of versioning in the model. The standard version allows the firm to retain consumers with a relatively low willingness to pay for capabilities, while the premium version enables the firm to charge a higher price to consumers who value additional capabilities more strongly. Versioning therefore allows the firm to segment the market more effectively and extract additional revenue from heterogeneous consumers compared with offering a single product.

4.4 NUMERICAL EXAMPLE

To illustrate the analytical solution, we use the same parameter values throughout the numerical analysis. These values are illustrative rather than estimated from observed data. They satisfy the maintained restrictions stated in Section 3.2 and provide a simple numerical example of the model.

The premium version, H, represents a highly capable AI model with relatively few safety restrictions, while the standard version, L, represents a safer model with lower capabilities that is available to all users.

$$c_H = 1,\ s_H = 0.2,\ c_L = 0.5,\ s_L = 0.8,\ \alpha = 0.8$$

These values capture the idea that the premium version is designed for users who value capabilities highly and are willing to accept fewer restrictions. The standard version offers stronger safety measures but provides lower capabilities. The parameter $\alpha = 0.8$ represents a relatively large alignment tax, meaning that consumers with a strong preference for capabilities place substantially less value on safety than consumers with lower values of θ.

Using the definition

$$q_i = c_i - \alpha s_i,$$

usable capabilities are

$$q_H = 1 - 0.8 * 0.2 = 0.84,$$
$$q_L = 0.5 - 0.8 * 0.8 = -0.14.$$

The example satisfies the maintained restrictions stated in Section 3.2. In particular, $q_L = -0.14 < 0 < q_H = 0.84, \; and \; s_L + q_L = 0.66 > 0$. The optimal menu derived below also has an interior cutoff, so the numerical solution lies in the fully covered regime analyzed in Section 4.2.

$$\Delta q = 0.98, \; \Delta s = 0.6, \; \Delta c = 0.5.$$

It also satisfies the conditions required for the single-version benchmarks in Proposition 1. For the standard version,

$s_L + 2q_L = 0.8 + 2 * (-0.14) = 0.52 > 0,$

so full market coverage is optimal. For the premium version,

$q_H = 0.84 > 0.20 = s_H,$

so the excluded-bottom solution applies.

Single-version benchmarks

Applying the results from Section 4.1 gives the benchmark outcomes.

Only the standard version

$$p_L^* = s_L + q_L = 0.8 - 0.14 = 0.66,$$
$$\pi_L^* = 0.66.$$

The standard version covers the entire market because every consumer purchases the product at the optimal price.

Only the premium version

$$p_H^* = \frac{q_H + s_H}{2} = \frac{0.84 + 0.20}{2} = 0.52,$$
$$\theta_0^* = \frac{q_H - s_H}{2q_H} = \frac{0.64}{1.68} = 0.381,$$
$$\pi_H^* = \frac{(q_H + s_H)^2}{4q_H} = \frac{1.04^2}{3.36} = 0.322.$$

Only 61.9% of consumers purchase the premium version, while consumers with $\theta < 0.381$choose the outside option.

The benchmark comparison illustrates the asymmetry identified in Section 4.1. The standard version generates higher profit than the premium version because serving the entire market more than offsets the higher margin obtained from premium users. The two-version menu must therefore outperform the standard-version benchmark in order to make versioning worthwhile.

Two-version menu

Applying the optimal cutoff from Section 4.2 gives

$$\theta_{HL}^{*} = \frac{\Delta s + q_H}{2\Delta q} = \frac{0.6 + 0.84}{1.96} = 0.735.$$

Consumers with $\theta < 0.735$ choose the standard version, while consumers above the cutoff choose the premium version. Substituting this cutoff into the pricing equations gives

$$\begin{aligned}
&p_L^* = 0.697,\\
&p_H^* = 0.817,\\
&n_L = 0.735,\ n_H = 0.265,
\end{aligned}$$

and

$$\pi^* = 0.697 \times 0.735 + 0.817 \times 0.265 = 0.729.$$

Compared with the benchmark cases, the menu combines the advantages of both strategies. The standard version retains consumers with lower values of θ, while the premium version generates additional revenue from consumers with a stronger preference for capabilities.

Figure 1 shows the consumer utility functions at the optimal menu under the numerical example. Both curves pass through zero at θ_{HL} = 0.735, confirming that the participation constraint binds at the interior cutoff rather than at the bottom of the distribution.

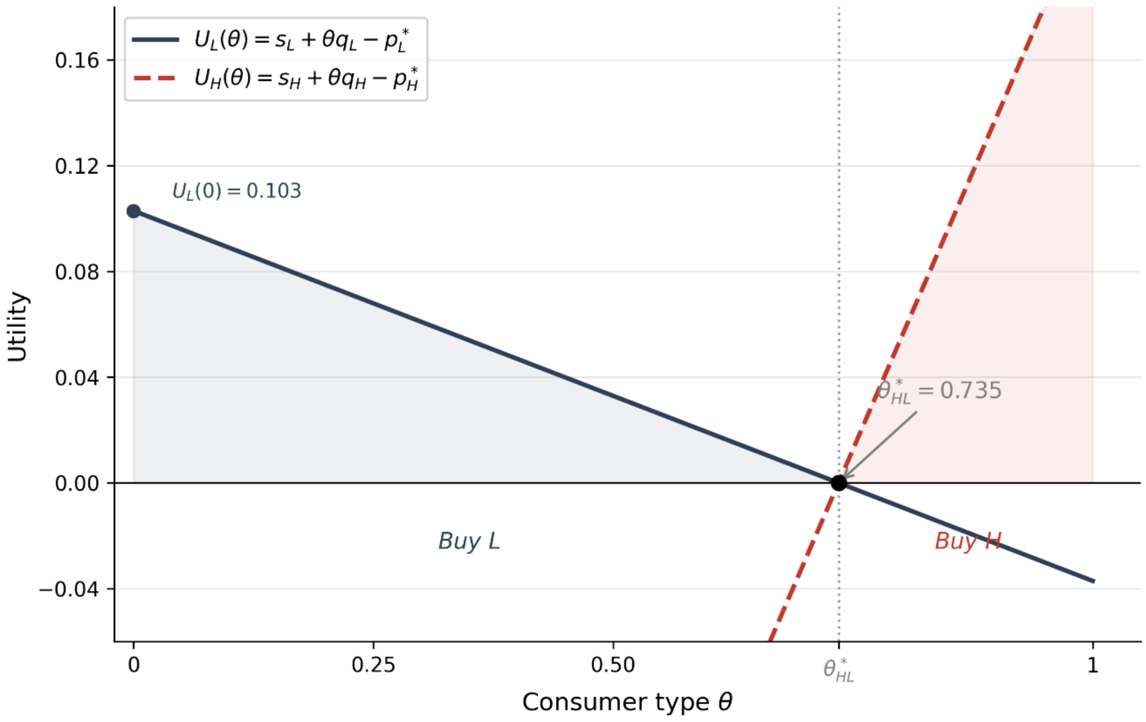

Figure 1: Consumer Utility at the Optimal Menu[3]

Comparison across strategies

The two-version menu generates the highest profit among the three pricing strategies. Relative to offering only the standard version, profit increases by 0.069 through the introduction of the premium segment. Relative to offering only the premium version, profit increases by 0.407 because the firm no longer excludes consumers with lower willingness to pay for capabilities.

The qualitative pattern illustrated here - a permissive premium tier alongside a more safety-oriented standard version, resembles xAI's positioning of Grok (xAI, 2023; xAI, 2024b; xAI, 2025). Under the illustrative parameter values used above, the model predicts that the firm would optimally divide consumers at θ = 0.735, with roughly three-quarters choosing the standard version. This is offered as a numerical illustration of the model's mechanics, not as an empirical estimate of Grok's actual customer split. Table 1 summarises the resulting prices, market coverage, profits, and ranking of the three pricing strategies.

[3] Both utility curves cross zero at θ_{HL}^*=0.735, where the participation constraint binds. Consumers with $\theta < \theta_{HL}^*$ purchase version L; those with $\theta > \theta_{HL}^*$ purchase version H. Figure 1 was produced using Python (matplotlib library) with assistance from Claude (Anthropic).

Strategy	p_L^*	p_H^*	Coverage	π*	Rank (by profit)
Only L	0.660	—	100%	0.660	2nd
Only H	—	0.520	61.9%	0.322	3rd
Menu (L + H)	0.697	0.817	100%	0.729	1st

Table 1: Comparison of pricing strategies under the illustrative example.

5. THIRD-PARTY HARM AND LIABILITY

Chapter 4 showed that offering both versions of the product is the firm's profit-maximizing strategy. By offering a safe standard version alongside a more capable and less restricted premium version, the firm can serve different groups of consumers and earn higher profit than by offering only one version.

This analysis, however, ignores an important feature of AI systems. The premium version is not only more capable than the standard version, but also less safe. As a result, some of the additional capabilities available to premium users can be used in ways that harm people outside the market. Examples include generating deepfakes, targeted harassment, or fraudulent content. These third parties are not involved in the transaction and receive no compensation for the harm they suffer. From the firm's perspective, these costs are external to the pricing decision made in Chapter 4.

To study this problem, the model is extended by introducing third-party harm and a liability rule. The firm is assumed to compensate a fraction of the harm generated by users of the premium version, while third parties continue to bear the remaining harm. This chapter examines how liability affects the firm's pricing decisions, market segmentation, and welfare, and assesses whether it is sufficient to bring the firm's incentives closer to the social optimum.

5.1 THIRD-PARTY HARM AND THE DAMAGE TECHNOLOGY

The analysis so far has focused only on consumers and the firm. In reality, however, AI use can also affect people not involved in the transaction. To capture this, the model introduces third-party harm.

Only the premium version H generates harm. This reflects the product configuration considered in the model: the premium version allows uses that the standard version's safety guardrails prevent. Users of version L therefore generate no harm outside the market.

Each user of version H generates an expected level of harm h. Let ρ_H denote the probability that a user of version H generates a harmful incident and let D_t denote the damage to third parties caused by such an incident. Expected harm per premium user is therefore

$$h = \rho_H D_t.$$

Since only users of the premium version generate harm, total expected harm depends on how many consumers purchase H. Because $n_H = 1 - \theta_{HL}$, total harm is:

$$\Gamma(\theta_{HL}) = h(1 - \theta_{HL}) = \rho_H D_t (1 - \theta_{HL}).$$

As the cutoff θ_{HL} increases, fewer consumers purchase the premium version and total harm falls. In this model, all third-party harm comes from the premium segment.

Two assumptions are important. First, third-party harm does not affect consumers' utility. Buyers do not bear the harm themselves and do not take it into account when choosing between the two versions, so consumer demand is exactly the same as in Chapter 4. Second, the firm cannot change the capabilities or safety of either version. The characteristics (c_i, s_i) are fixed, so the firm's only choice is how many consumers receive the premium version through the cutoff θ_{HL}.

5.2 LIABILITY AND THE FIRM'S PROBLEM

Third-party harm creates a difference between what is best for the firm and what is best for society. Without liability, the firm ignores the harm generated by the premium version and chooses θ_{HL}only to maximize profit. To account for this, the model introduces a strict liability rule under which the firm compensates a share $\lambda \in [0, 1]$ of total third-party harm.

When $\lambda = 0$, the model is the same as in Chapter 4. When $\lambda = 1$, the firm bears the full cost of the harm generated by users of the premium version.

The liability payment $\lambda\Gamma$, enters the firm's profit function as an additional cost. Given the cutoff θ_{HL}, profit is

$$\boldsymbol{\pi(\theta_{HL}) = p_L n_L + p_H n_H - \lambda h(1 - \theta_{HL}).}$$

Consumer demand is unchanged. Buyers do not bear third-party harm and therefore do not take it into account when choosing between the two versions. The pricing equations are therefore the same as in Chapter 4:

$$p_L(\theta_{HL}) = s_L + \theta_{HL} q_L,$$
$$p_H(\theta_{HL}) = s_H + \theta_{HL} q_H.$$

Substituting these expressions into the profit function and maximizing with respect to θ_{HL}gives the following result.

Proposition 4 (Effect of Liability on the Firm's Choice): Suppose $\boldsymbol{\Delta s + q_H + \lambda h < 2\Delta q}$**.** Then the firm's optimal interior cutoff under liability is

$$\boldsymbol{\theta_{HL}^*(\lambda) = \frac{\Delta s + q_H + \lambda h}{2\Delta q}}.$$

If $\boldsymbol{\Delta s + q_H + \lambda h \geq 2\Delta q}$, the interior solution reaches the upper boundary and version H is no longer offered.

Substituting the pricing equations into the profit function gives

$$\pi(\theta_{HL}) = (s_H - \lambda h) + (\Delta s + q_H + \lambda h)\theta_{HL} + (q_L - q_H)\theta_{HL}^2.$$

The first-order condition $d\pi/d\theta_{HL} = 0$ gives the stated expression. The second-order condition $2(q_L - q_H) = -2\Delta q < 0$ confirms that this solution is a maximum.

As liability increases, the firm assigns fewer consumers to the premium version. The optimal cutoff θ_{HL}^* increases, the premium segment becomes smaller, and total third-party harm falls. The market remains fully covered, although sufficiently high liability may eliminate the premium segment entirely.

Prices also change. Since $q_H > 0$, a higher cutoff increases the price of the premium version because only consumers with higher values of θ remain in that segment. Since $q_L < 0$, the price of the standard version falls because it is now offered to consumers who value it less. Liability therefore does not exclude consumers from the market. Instead, it shifts some consumers from the premium version to the standard version.

5.3 SOCIAL WELFARE AND THE LIMITS OF LIABILITY

Social welfare in this model consists of consumer surplus, firm revenue, and third-party welfare. We first define consumer surplus and firm revenue for a given cutoff θ_{HL}.

Aggregate consumer surplus is

$$CS(\theta_{HL}) = \int \int_{0}^{\theta_{HL}} U_L(\theta)d\theta + \int_{\theta_{HL}}^{1} U_H(\theta)d\theta.$$

Using the prices associated with the cutoff, this becomes:

$$\boldsymbol{CS(\theta_{HL}) = -\frac{1}{2}q_L\theta^2{}_{HL} + \frac{1}{2}q_H(1-\theta_{HL})^2.}$$

Firm revenue is

$$\boldsymbol{R(\theta_{HL}) = \ p_L\theta_{HL} + p_H(1-\theta_{HL}).}$$

The distinction between revenue and profit is important for the welfare analysis. The liability payment λΓ is a transfer from the firm to the affected third parties rather than a loss of resources to society. It therefore cancels out when social welfare is calculated. For this reason, firm revenue R, rather than profit net of liability, enters the welfare function.

Social welfare is therefore:

$$\boldsymbol{W(\theta_{HL}) = CS(\theta_{HL}) + R(\theta_{HL}) - \Gamma(\theta_{HL}).}$$

Although the liability share λ does not appear directly in the welfare function, it still affects welfare through the firm's behavior. A higher value of λ changes the firm's optimal cutoff, which in turn affects consumer surplus, firm revenue, and third-party harm.

The firm's objective is to maximize profit, while a social planner seeks to maximize overall welfare. Because the planner accounts for consumer surplus and the full social cost of third-party harm, the planner's preferred allocation need not coincide with the firm's.

To clarify this comparison, the planner does not choose product characteristics or prices. Product characteristics remain fixed, while the firm continues to choose prices. The planner chooses the allocation of consumers between the two versions, represented by the cutoff θ_{HL}, taking into account both consumer utility and third-party harm. The planner's allocation is therefore used as a benchmark for comparison with the firm's allocation.

This comparison highlights two possible sources of the difference between the private and the socially preferred allocation. The first is the externality distortion, which arises because the firm internalizes only a fraction λ of the harm generated by users of the premium version. The second is the monopoly distortion, which arises because the firm maximizes profit rather than social welfare and therefore does not internalize consumer surplus. The numerical analysis in the next section examines the importance of these two distortions and evaluates whether liability alone is sufficient to bring the firm's choice closer to the social optimum.

5.4 NUMERICAL ILLUSTRATION

The model is now illustrated using the parameter values introduced in Chapter 4 together with the third-party harm introduced in Chapter 5. This section shows how different levels of liability affect the firm's decisions and compares the firm's outcome with the socially preferred one.

5.4.1 Parameter Values and Harm Technology

The parameter values are the same as in Chapter 4:

$$\boldsymbol{c_H = 1,\ s_H = 0.2,\ c_L = 0.5,\ s_L = 0.8,\ \alpha = 0.8,}$$

which give

$$q_H = 0.84,\ q_L = -0.14,\ \Delta q = 0.98,\ \Delta s = 0.6.$$

The harm technology follows the assumptions introduced in Section 5.1. For the numerical illustration, the baseline probability of harmful use is $\rho_0 = 0.2$. Since the premium version has safety level $s_H = 0.2$, the probability that a user of version H causes a harmful incident is

$$\rho_H = \rho_0(1 - s_H) = 0.2 \times 0.8 = 0.16.$$

The standard version blocks harmful uses, so

$$\rho_L = 0.$$

We set the damage from each incident to $D_t = 2$. Expected harm per premium user is therefore:

$$h = \rho_H D_t = 0.16 \times 2 = 0.32.$$

and total expected harm is

$$\Gamma(\theta_{HL}) = 0.32(1 - \theta_{HL}).$$

This means that each additional consumer who buys the premium version increases expected third-party harm by 0.32.

5.4.2 The firm's response to liability

Applying Proposition 4 gives the firm's optimal cutoff under liability:

$$\theta^*_{HL}(\lambda) = \frac{1.44 + 0.32\lambda}{1.96}$$

The table below reports the firm's optimal cutoff, market shares, prices, profit, consumer surplus, third-party harm, and social welfare for three values of the liability share.

λ	θ*_HL	n*_H	p*_L	p*_H	π	CS	Γ	W

0	0.7347	0.2653	0.6971	0.8171	0.7290	0.0674	0.0849	0.7115
1/2	0.8163	0.1837	0.6857	0.8857	0.6930	0.0610	0.0588	0.7246
1	0.8980	0.1020	0.6743	0.9543	0.6702	0.0610	0.0326	0.7312

Table 2: Optimal firm behaviour under liability

The results are consistent with the analytical model. As liability increases, the firm assigns fewer consumers to the premium version. The premium segment becomes smaller, while the standard version is offered to a larger share of consumers. At the same time, the price of the premium version increases and the price of the standard version decreases, as shown in Section 5.2.

Third-party harm falls as liability increases because fewer consumers purchase the premium version. Social welfare increases across the three liability levels, reflecting the reduction in harm. The market remains fully covered in all cases.

For comparison, the table below reports the outcomes for the two single-version benchmarks together with the menu when there is no liability.

Regime	p*_L	p*_H	Coverage	π	Γ	W
Only L	0.660	—	100%	0.660	0.000	0.730
Only H	—	0.520	61.9%	0.322	0.198	0.285
Menu (λ = 0)	0.697	0.817	100%	0.729	0.085	0.712

Table 3: Profit, harm and welfare rankings across pricing strategies (λ = 0).

The comparison highlights the difference between profit and welfare. The menu generates the highest profit, but selling only the standard version produces higher welfare because it generates no third-party harm. Without liability, the firm chooses the menu because it maximizes profit, even though it is not the welfare-maximizing outcome.

5.4.3 The planner's optimum and the limits of liability

The firm's optimal cutoff under full liability is $\theta_{HL}^{*}(1) = 0.8980$. To assess whether this outcome is socially optimal, we derive the cutoff that maximizes social welfare,

$$W(\theta_{HL}) = CS(\theta_{HL}) + R(\theta_{HL}) - \Gamma(\theta_{HL}).$$

Using the expression for consumer surplus derived in Section 5.3,

$$CS(\theta_{HL}) = \frac{1}{2} \cdot 0.14 \cdot \theta_{HL}^2 + \frac{1}{2} \cdot 0.84 \cdot (1 - \theta_{HL})^2.$$

and substituting this expression together with

$$R(\theta_{HL}) = 0.2 + 1.44\theta_{HL} - 0.98\theta_{HL}^2$$

and

$$\Gamma(\theta_{HL}) = 0.32(1 - \theta_{HL})$$

into the welfare function gives

$$\boldsymbol{W(\theta_{HL}) = 0.30 + 0.92\theta_{HL} - 0.49\theta_{HL}^2}.$$

The first-order condition,

$$\frac{dW}{d\theta_{HL}} = 0,$$

gives

$$\boldsymbol{\theta_{HL}^{W} = \frac{0.92}{0.98} = 0.9388}.$$

The planner would assign 93.9% of consumers to the standard version, compared with 89.8% under full liability. Even under full liability, the firm's choice differs from the planner's preferred allocation.

To examine whether this difference is driven only by third-party harm, we set $h = 0$ and repeat the analysis. With no harm, welfare becomes

$$W(\theta_{HL}) = CS(\theta_{HL}) + R(\theta_{HL}),$$

and the first-order condition gives

$$\theta_{HL}^{W}(h = 0) = \frac{0.60}{0.98} = 0.6122.$$

The firm, however, chooses

$$\theta_{HL}^{*}(0) = 0.7347.$$

Even when there is no third-party harm, the firm's cutoff differs from the planner's. This shows that part of the gap is unrelated to the externality. It arises because the firm maximises profit, while the planner also takes consumer surplus into account.

The liability share that would induce the firm to choose the planner's cutoff solves

$$\frac{1.44 + 0.32\lambda^{*}}{1.96} = 0.9388,$$

which gives

$$\boldsymbol{\lambda^{*} = 1.25.}$$

Since $\lambda^{*} > 1$, no feasible liability rule can induce the firm to choose the socially optimal allocation. Full liability internalizes the entire externality, but it does not eliminate the gap between the firm's choice and the planner's optimum because the monopoly distortion remains.

Finally, there exists a liability share above which the two-version menu generates higher welfare than offering only the standard version. Setting

$$W(\text{menu}, \lambda) = W(\text{only } L) = 0.730$$

gives

$$\lambda^{**} = 0.875.$$

Since θ_{HL} (λ) increases linearly in λ and the welfare function W(θ_{HL}) is increasing in θ_{HL} for all θ_{HL}< θ_{HL}^{W}= 0.9388, welfare under the menu is monotonically increasing in λ over the feasible range [0, 1]. In particular, even under full liability, θ_{HL} (1) = 0.8980 < θ_{HL}^{W}. The threshold λ** = 0.875 is therefore the unique value at which W(menu, λ) = W(only L), and the menu generates higher welfare for λ > λ**.

This suggests that versioning along the safety dimension is not necessarily welfare-reducing. Its welfare effects depend on the liability regime under which the firm operates.

The comparison between the firm's choice and the planner's optimum reveals two separate sources of distortion. The first is the externality distortion, which arises because the firm internalizes only a fraction of third-party harm. The second is the monopoly distortion, which remains even when third-party harm is absent because the firm maximizes profit rather than total welfare.

In the baseline example, the liability share that would induce the firm to choose the planner's preferred cutoff is $\lambda^* = 1.25$. Since this value exceeds one, no feasible liability rule can implement the planner's allocation in this case. Full liability internalizes the externality but does not eliminate the remaining monopoly distortion.

5.4.4 SENSITIVITY TO HARM SEVERITY

So far, the numerical analysis has assumed $D_t = 2$. To examine how the results depend on the severity of harm, we vary $D_t$while keeping $\rho_0 = 0.2$ fixed. Since $h = \rho_H \cdot D_t$ and $\rho_H = 0.16$, the harm generated by each premium user becomes

$$h = 0.16 \cdot D_t.$$

The firm's choice without liability does not depend on D_t. When $\lambda = 0$, harm does not enter the firm's optimization problem, so the optimal cutoff remains

$$\theta^*_{HL} = 0.7347.$$

The planner's preferred cutoff, however, changes with the severity of harm. It is given by

$$\theta^W_{HL} = \frac{\Delta s + h}{\Delta q} = \frac{0.6 + 0.16D_t}{0.98},$$

while the liability share that would induce the firm to choose the planner's allocation is

$$\frac{1.44 + 0.16D_t\lambda^*}{1.96} = \theta^W_{HL},$$

which gives

$$\lambda^* = 2 - \frac{1.5}{D_t}.$$

The results are reported in the next table.

D_t	h	θ*_HL(0)	θ^W_HL	λ*	Policy conclusion
0.50	0.08	0.7347	0.6939	−1.00	Harm too low; monopoly distortion dominates; liability not warranted
0.75	0.12	0.7347	0.7347	0.00	Firm and planner agree; no liability needed
1.00	0.16	0.7347	0.7755	0.50	Moderate liability achieves social optimum
1.50	0.24	0.7347	0.8571	1.00	Full liability exactly achieves social optimum
2.00	0.32	0.7347	0.9388	1.25	Full liability insufficient; complementary instruments needed
2.50	0.40	0.7347	1.00	1.30	Planner would eliminate H entirely; liability far from sufficient

Table 4: Sensitivity of results to harm severity D_t (ρ_0 = 0.2 fixed).

When $D_t = 0.50$, the efficient liability share is negative. In this case, the externality is too small to outweigh the monopoly distortion. The firm already assigns fewer consumers to the premium version than the planner would prefer, so introducing liability would move the firm even further away from the social optimum.

When $D_t = 0.75$, the two distortions exactly offset each other. The planner chooses the same cutoff as the unregulated firm, so no liability is needed to achieve the socially preferred allocation.

When $D_t = 2.50$, the planner's unconstrained optimum exceeds one. Since the cutoff is restricted to the interval $[0, 1]$, the planner chooses $\theta_{HL}^W = 1$, assigning all consumers to the standard version and eliminating the premium version. The liability share that would induce the firm to make the same choice is found by setting $\theta_{HL}^*(\lambda) = 1$, which gives $\lambda^* = 1.30$. Since this value exceeds the feasible maximum of one, liability alone cannot induce the firm to eliminate version H.

The case $D_t = 0.50$ is also useful because it illustrates how the conclusions depend on the size of the externality. When harm is relatively small, the monopoly distortion is more important than the externality. For this reason, Chapter 5 uses $D_t = 2$ as the baseline example, where the externality is large enough for liability to play a meaningful role.

6. CASE STUDY – GROK AND XAI AS AN ILLUSTRATION OF THE MODEL

The theoretical analysis developed in the previous chapters considers AI versioning when consumers differ in the relative value they place on capability and safety. It also shows how a monopolist prices two given versions when the premium version has higher capability but lower safety, and when the use of that version can create harm for people outside the transaction.

This chapter examines whether some of these mechanisms can be observed in an actual AI market. The case of Grok, developed by xAI, is used as an illustrative case.

The purpose is not to estimate the model or to claim that Grok follows the exact structure assumed in the theoretical analysis. The information required to estimate the model, such as consumer preferences, the value users place on safety, the probability of harmful use and the firm's expected liability costs, is not publicly available. The case study therefore has a more limited objective. It asks whether the main economic mechanisms identified in the model can be recognised in the way xAI has developed and differentiated access to Grok.

This distinction is important for the interpretation of the results. The numerical values obtained in Chapters 4 and 5 are generated by the model and are not estimates of Grok's actual market. Grok is used to provide an empirical context in which the theoretical mechanisms can be discussed.

6.1 GROK AS AN EXAMPLE OF AI VERSIONING

Grok was introduced by xAI in November 2023. From the beginning, xAI positioned Grok as an AI system intended to answer a broad range of questions, including some questions that other AI systems might reject (xAI, 2023) .

The product subsequently developed beyond a single chatbot. In August 2024, xAI introduced Grok 2 and Grok 2 mini (xAI, 2024a). Later in 2024, Grok was made available to all users on X, while the company's paid subscription structure continued to provide differences in access and usage (xAI, 2024b).

The differentiation became more pronounced as new models were introduced. In July 2025, xAI introduced Grok 4 and made it available to SuperGrok and Premium+ subscribers, while also introducing a SuperGrok Heavy tier with access to Grok 4 Heavy (xAI, 2025).

This development is relevant to the versioning model used in this thesis. The theoretical model contains only two versions, standard and premium, while the actual Grok product has a more complicated structure. Nevertheless, the basic economic idea is similar. A broad group of users receives standard access, while users willing to pay receive additional capabilities, higher usage limits or access to more advanced models.

The two-version model should therefore be understood as a simplification of the actual product menu. Its purpose is to isolate the economic mechanism rather than reproduce every subscription level offered by xAI.

6.2 CAPABILITY AND SAFETY

The distinction between capability and safety is more difficult to observe directly than the distinction between free and paid access.

Capability can be observed through the introduction of increasingly advanced Grok models and additional functions. Safety, however, is not a single measurable characteristic. It can involve content filtering, refusal behavior, restrictions on image generation, restrictions on editing, and other forms of access control.

This creates an important difference between the theoretical model and the real product. In the model, safety is represented as a single product characteristic. In practice, safety is multidimensional and can differ across functions.

For the purposes of this case study, the safety dimension is therefore interpreted more broadly as the degree to which the system restricts potentially harmful outputs. The case does not attempt to assign a numerical safety value to Grok.

This is also why the Grok case should not be presented as proof that the premium version is objectively less safe in every respect. The more limited claim is that differences in the degree of restriction can form part of product differentiation, which is consistent with the type of trade-off considered in the theoretical model.

6.3 THIRD-PARTY HARM

The third-party harm introduced in Chapter 5 is central to the comparison with Grok.

The model assumes that some uses of the premium product can impose costs on individuals who are not part of the transaction between the firm and its users. In such a situation, the user and the firm may consider the private benefits and costs of using the product, while part of the social cost falls on third parties.

The Grok case provides a concrete example of why this mechanism is relevant. In January 2026, the California Attorney General announced an investigation into the production and distribution of non-consensual sexually explicit material generated using Grok (California Department of Justice, 2026a). The investigation concerned material depicting women and children and the use of such material to harass individuals online.

Two days later, on January 16, 2026, the California Attorney General sent xAI a cease-and-desist letter demanding that the company stop the creation and distribution of deepfake non-consensual intimate images and child sexual abuse material (California Department of Justice, 2026b).

These events cannot be used to estimate the harm parameter in the theoretical model. They nevertheless illustrate the type of externality considered in Chapter 5. The people affected by harmful generated content may not be users of the product and may have no contractual relationship with the firm.

This distinction between private and social costs is important for the welfare analysis. It explains why a product can be privately valuable to the user while simultaneously creating a cost that is not reflected in the transaction price.

6.4 INTERPRETING THE NUMERICAL RESULTS

The numerical results in Chapters 4 and 5 are illustrative and should be kept separate from the Grok evidence. Under the baseline example in Chapter 4, selling only the standard version produces a profit of $\pi_L^* = 0.660$, while selling only the premium version produces a profit of $\pi_H^* = 0.322$. When both versions are offered, the firm's profit increases to $\pi^* = 0.729$. Under the assumptions of the numerical example, the firm therefore benefits from offering a menu rather than selling either version alone. The two versions allow consumers with different preferences to sort themselves between the products. The optimal cutoff is approximately $\theta_{HL}^* = 0.735$. Under the assumed uniform distribution of consumers, this means that approximately 26.5% of consumers choose the premium version, while the remaining 73.5% choose the standard version. The 26.5% figure should not be interpreted as an estimate of the proportion of Grok users who pay for premium access. Such a comparison is not possible with the available data. Instead, the result illustrates the type of segmentation that can arise when consumers differ in their relative valuation of capability and safety. The Grok market provides a qualitative setting that is compatible with this mechanism. xAI has maintained broad access to Grok while developing paid tiers with additional capabilities, higher usage limits and access to more advanced models (xAI, 2024b; xAI, 2025). The similarity should not be overstated. The actual Grok market has several subscription levels, and the number and characteristics of users in each group are not publicly available in sufficient detail. The case therefore illustrates the logic of segmentation, but not of the numerical value of the model's cutoff.

6.5 LIABILITY AND FIRM’S RESPONSE

Chapter 5 introduces liability as a cost borne by the firm when premium users generate third-party harm. In the theoretical model, a higher liability share increases the private cost associated with the premium segment and raises the firm's optimal cutoff. As a result, fewer consumers purchase the premium version.

The Grok case cannot provide a direct empirical test of this mechanism because the regulatory pressure surrounding Grok is not equivalent to the liability parameter used in the model. The firm's costs may include legal exposure, regulatory restrictions, reputational effects, and changes in access conditions, none of which can be separated using publicly available information.

Following the January 2026 controversy, xAI initially restricted image generation and editing for free users, while some of these functions remained available to paying subscribers (Iyer, 2026). Restrictions on the sexualised editing of real people were subsequently applied more broadly (Milmo, 2026). The initial response therefore operated partly through access conditions rather than through the complete removal of the underlying capability.

This provides a useful comparison with the mechanism in the model. An increase in the private cost associated with the premium version does not necessarily require the firm to remove that version. The firm may instead reduce access to it by changing prices or access conditions. If the premium segment remains profitable, differentiated access can remain part of the firm's strategy.

The numerical results in Chapter 5 help to interpret this mechanism. In the baseline example, the firm's cutoff increases from

$$\theta_{HL}^{*}(0) = 0.7347$$

without liability to

$$\theta_{HL}^{*}(1) = 0.8980$$

under full liability. The premium segment therefore falls from approximately 26.5% of consumers to 10.2%.

The planner's preferred cutoff is higher still:

$${\theta_{HL}^{*}}^{W} = 0.9388$$

These numbers should not be mapped directly onto Grok users or actual regulatory measures. Their purpose is to show the direction of the mechanism. As the private cost associated with harmful premium use increases, the premium segment becomes smaller. At the same time, the numerical analysis shows that even full liability does not reproduce the planner's preferred allocation in the baseline example.

The sensitivity analysis also cautions against interpreting this result as an argument that stronger liability is always preferable. The effect of liability depends on the severity of third-party harm. When harm is relatively small, the monopoly distortion can dominate the externality distortion, so increasing liability need not move the allocation closer to the social optimum. The severity of third-party harm cannot be estimated from the Grok evidence considered here.

The Grok case therefore provides a qualitative interpretation of the mechanism rather than evidence for a particular liability value. Regulatory and legal pressure can change access conditions without necessarily eliminating the commercial incentive to maintain differentiated products.

6.6 DISCUSSION

The Grok case shares several features with the theoretical setting, but it also shows where the model simplifies the real market.

The first similarity is product differentiation. xAI has developed different levels of access to Grok and has introduced increasingly advanced models and subscription tiers (xAI, 2024b; xAI, 2025). This resembles the basic versioning structure used in the model, even though the real product menu contains more than two alternatives.

The second similarity concerns the distinction between capability and safety. More advanced AI capabilities can be offered together with different restrictions on how those capabilities are accessed or used. The Grok case therefore illustrates why capability and safety can be treated as separate product characteristics, even though their relationship in practice is more complicated than the model assumes.

The third similarity is the possibility of third-party harm. The January 2026 intervention by the California Attorney General provides an example of harmful use of an AI image generation function affecting individuals who are not part of the original transaction (California Department of Justice, 2026a, 2026b). This is the type of external cost represented by the harm term in Chapter 5.

At the same time, the case shows why the theoretical model should not be interpreted too literally. Grok has more than two product levels, safety is multidimensional, consumer preferences are more complex than the single parameter used in the model, and the firm's private cost of harmful use cannot be observed directly.

The Grok case should therefore not be interpreted as a validation of the numerical example. Its value is instead qualitative. Grok provides a real setting in which capability, access, safety, and third-party harm interact in ways that resemble the mechanisms considered in the model. In particular, the case shows that a response to harmful use can take the form of changes in access conditions rather than the complete removal of a capability. This gives the theoretical cutoff mechanism a useful real-world interpretation while keeping a clear distinction between the model's numerical results and the available empirical evidence.

7. CONCLUSION

This thesis examined how a monopolist prices AI products when consumers differ in the relative value they place on capability and safety. The analysis considered two versions of an AI system. The premium version has higher capability but lower safety, while the standard version has lower capability and higher safety. The characteristics of the two products are taken as given, so the firm's decision concerns prices and the resulting allocation of consumers between the two versions.

The analysis builds on the standard economic logic of versioning and second-degree price discrimination. The model extends this framework by allowing the two products to differ in both capability and safety, while preserving one-dimensional consumer sorting. Consumers differ in how strongly they value capability relative to safety, so the version with higher capability is not necessarily preferred by every consumer. This type of sorting is particularly relevant for AI products, where greater capability can be offered together with different levels of restrictions and safeguards.

The single-version benchmarks show that the two products serve different parts of the market. Under the maintained parameter restrictions, utility from the standard version decreases with consumer type, while utility from the premium version increases. When the standard version is sold alone, the optimal benchmark serves the entire market. When only the premium version is offered, consumers with sufficiently low valuations of capability choose the outside option.

The pricing analysis shows that versioning can increase the firm's profit. Under the numerical example used in the thesis, the standard-only strategy generates a profit of 0.660, while the premium-only strategy generates 0.322. The two-version menu generates a higher profit of 0.729. The firm benefits from allowing consumers to sort themselves between the two products rather than making a single version attractive to the entire market.

The optimal cutoff in this numerical example is approximately $\theta^* = 0.735$. Given the assumed uniform distribution of consumer types, approximately 26.5% of consumers choose the premium version, while the remaining consumers choose the standard version. These values are generated by the example and should not be interpreted as estimates of an actual AI market. Their purpose is to illustrate the model's sorting mechanism.

The welfare analysis changes the problem once the premium version harms third parties. Without liability, this harm does not enter the firm's private optimization problem. The firm accounts for the profitability of the premium segment without bearing the full costs imposed on people outside the transaction. Liability changes this incentive by making the firm responsible for a share of the harm generated by premium users.

As the liability share increases, the optimal cutoff rises and the premium segment becomes smaller. In the baseline example, the firm's cutoff increases from 0.7347 without liability to 0.8980 under full liability. Consequently, the share of consumers using the premium version falls from approximately 26.5% to 10.2%. Liability therefore reduces third-party harm by changing the firm's allocation of consumers between the two versions.

However, the firm's allocation does not necessarily coincide with the social optimum. In the baseline case, the planner's preferred cutoff is 0.9388, which is higher than the firm's cutoff even under full liability. The liability share that would induce the firm to reproduce this allocation is $\lambda * = 1.25$, which lies outside the feasible range. This is because two different distortions are present. The first comes from the externality created by third-party harm. The second comes from monopoly pricing and exists even without external harm. Liability directly addresses the first distortion, but it does not remove the second.

This distinction is particularly important in the sensitivity analysis. When third-party harm is relatively small, the monopoly distortion can dominate the externality distortion. In that case, the monopolist already allocates too few consumers to the premium version relative to the social planner, and additional liability can move the allocation further away from the social optimum. The analysis also identifies an intermediate case in which the two distortions offset each other, so that the unregulated firm's allocation coincides with the planner's preferred allocation. As harm becomes larger, the externality becomes more important, and liability becomes more useful.

The results therefore do not support a simple conclusion that stronger liability is always better. Its effect depends on the severity of third-party harm and on the distortion already created by monopoly pricing. Regulation that is appropriate when the externality is large may not have the same welfare effect when the expected harm is small.

The Grok case provides a qualitative illustration of these mechanisms. xAI has developed Grok through several models and subscription levels, with differences in access, capabilities, and restrictions (xAI, 2024a, 2024b; xAI, 2025). The January 2026 controversy surrounding non-consensual sexually explicit material generated using Grok also illustrates how the use of an AI system can impose costs on individuals outside the transaction between the provider and the user (California Department of Justice, 2026a, 2026b).

The case does not provide an empirical test of the model. Grok has more than two product levels, safety is multidimensional, and the available evidence does not allow the model's consumer shares, harm parameter or liability parameter to be estimated. Instead, it shows how product differentiation, access restrictions, and third-party harm can interact in an actual AI market. The response to harmful use also illustrates that firms may change access conditions rather than eliminate a capability entirely, which provides a useful real-world interpretation of the allocation mechanism considered in the model (Iyer, 2026; Milmo, 2026).

Several limitations remain. First, the model takes capability and safety as given. A natural extension would be to make product characteristics endogenous and allow the firm to choose capability and safety before setting prices. This would make it possible to study whether liability changes not only consumer allocation but also product design.

Second, the analysis considers a monopolist, while AI providers operate in an increasingly competitive market. Competition could affect prices, product differentiation and incentives to invest in safety. Extending the framework to competing AI providers could therefore help connect the versioning mechanism studied here with the broader literature on competition and AI safety.

Third, the numerical analysis is illustrative rather than empirical. Better data on consumer preferences, subscription choices, harmful uses, and third-party costs would make it possible to estimate some of the parameters of the model and examine whether the predicted patterns of market segmentation are observed in practice.

Despite these limitations, the model provides a useful framework for thinking about the relationship between AI pricing and safety. The main conclusion is not that premium AI products are necessarily harmful, nor that stronger liability is always desirable. Rather, the analysis shows that pricing, product differentiation, access, and safety can be closely connected. When a firm determines who receives access to a more capable but less safe version, its pricing decision also affects the distribution of potential benefits and harms. The thesis's main contribution is to connect these issues within a single versioning framework. The pricing analysis shows how a monopolist can profit from separating consumers according to their preferences for capability and safety. The welfare analysis then shows why the resulting allocation may differ from the one preferred by a social planner once both consumer surplus and third-party harm are taken into account. Liability can reduce this difference in some cases, but its effectiveness depends on the interaction between the externality and the distortion created by monopoly pricing.

The Grok case suggests that these questions have practical relevance beyond the theoretical model. Current AI markets already combine different levels of capability, access and restriction, while the consequences of harmful AI use can extend beyond the consumers who use the system. Future research could extend this framework by studying endogenous product design, competition between AI providers and the interaction between pricing, access restrictions and liability.

Declaration on the Use of Artificial Intelligence Tools

I declare that this thesis is my own work and that I take full responsibility for its content. During the preparation of the thesis, I used Claude (Anthropic) as an AI-assisted tool for the following purposes:

- **Language assistance:** As a non-native English speaker, I used the tool to check grammar, improve phrasing, and make parts of the text clearer. I reviewed and approved the final wording used in the thesis.

- **Figure generation:** Figure 1 was produced in Python using matplotlib with AI assistance, as I do not have programming experience. The figure is based on the model and parameter values used in the thesis.

- **Calculation and consistency checks:** I used the tool to check calculations and the consistency of numerical results across different sections of the thesis.

- **Literature search support:** I used the tool to identify potentially relevant academic papers and to obtain initial summaries of their main arguments. The sources cited in the thesis were subsequently read and checked independently.

I take full responsibility for the economic model, theoretical arguments, analytical results, interpretation of the findings, selection and use of sources, and the final content of the thesis.

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